SAMPADA JOSHI

+91 9742174604 | joshisampada11115@gmail.com | github.com/sampada115 | linkedin.com/in/sampada-joshi

EDUCATION

B.Tech - Computer Science and Engineering, PES University, Bengaluru 2023 – 2027 CGPA: 8.76/10

SKILLS

Languages: C, C++, Python, R, JavaScript, SQL **Technologies:** React.js, Node.js, Express.js, MySQL, MongoDB, REST APIs **ML/AI:** Scikit-learn, Pandas, NumPy, Matplotlib, TensorFlow, PyTorch, Hugging Face Transformers, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG) **Tools:** Git, GitHub, VS Code, Docker **Coursework:** OOP, DBMS, OS, Computer Networks, Machine Learning

EXPERIENCE

Research Intern, Center for Computer Networks and Cyber Security (CCNCS), PES University June 2025 – July 2025

- Engineered SecurePrompt, a multi-layered defense architecture (Prompt Validation + RAG-based Response Validation via FAISS) for detecting and mitigating prompt injection attacks in finance-domain LLM applications, combining DistilBERT-based ML classification with rule-based filtering.
- Benchmarked DistilBERT against BERT, DIET, and RoBERTa (90% accuracy, 88.89% F1); extended to a 120,000-prompt multi-domain dataset achieving 99.68% overall accuracy. Co-authored corresponding paper, published at IEEE ICCCMLA 2025.

PROJECTS

SciGuard – Scientific Image Forgery Detection

- Designed a dual-stream model (ResNet-50 RGB stream + SRM-guided Swin Transformer noise stream) for forgery detection and localization in scientific images; achieved F1 0.4852 on RECOD-AI LUC dataset, a +8.4% improvement over an RGB-only baseline, with interpretable heatmaps.

Coral Reef Bleaching Risk Prediction

- Built a multimodal fusion pipeline combining NOAA satellite thermal stress indicators (SST, HotSpot, DHW, SSTA) with SciBERT text embeddings from bleaching reports for region- and time-aware risk prediction across the Great Barrier Reef.

GenAI-Based Temporal Safety & Threat Detection for Software-Defined Vehicles

- Contributed to an agentic GenAI framework (LLM scenario generation via Groq/Llama-3.3-70B, RAG retrieval, Q-learning RL) for temporal safety validation and cyber-threat detection in SDVs per ISO 26262/21434; achieved 91% event chain accuracy and 88% threat detection rate. Co-authored IEEE paper *(Under Review)*.

PUBLICATIONS

SecurePrompt: Detecting and Mitigating Prompt Attacks – IEEE ICCCMLA 2025

CERTIFICATIONS

DeepLearning.AI – Machine Learning Specialization Deloitte – Data Analytics Job Simulation HackerRank – Problem Solving (Basic)